BIOBUZZ Autonomous Routes

One page per autonomous OpMode, drawn from the JSON route files the robot runs and the Pedro paths it follows. Points and times are estimates; the checks catch rule and field problems before the robot does.

OPMODE	ROUTE FILE	EST. POINTS	EST. TIME	CHECKS
Red: Leave + Park	leave_and_park.json	**8 pts**	4.7 s	**1 problem (see page)**
Blue: Leave + Park	leave_and_park.json	**8 pts**	4.7 s	**1 problem (see page)**
Red: My Route	my_route.json	**8 pts**	4.7 s	**1 problem (see page)**
Blue: My Route	my_route.json	**8 pts**	4.7 s	**1 problem (see page)**
Red: Vision Seek (test)	vision_seek.json	**0 pts**	9.4 s	**3 problems (see page)**
Blue: Vision Seek (test)	vision_seek.json	**0 pts**	9.4 s	**3 problems (see page)**

HOW TO READ A PAGE

- Path the robot follows, in step order; the chevron points the way it drives.
- Robot at START (18 x 18 in max, R102); the thick edge is its front.
- Robot after each path.
- Robot at the end of AUTO, where LEAVE and PARK are judged.
- 2 — Routine step number, matching the list on the page.
- Waypoint the robot passes through; hollow means a Bezier control point it doesn't reach.

WHAT THE NUMBERS ASSUME

Field: docs/coordinates.md. 144 in square, origin at the red GARDEN corner, x toward the blue wall, y away from the audience, heading 0° = facing +x.

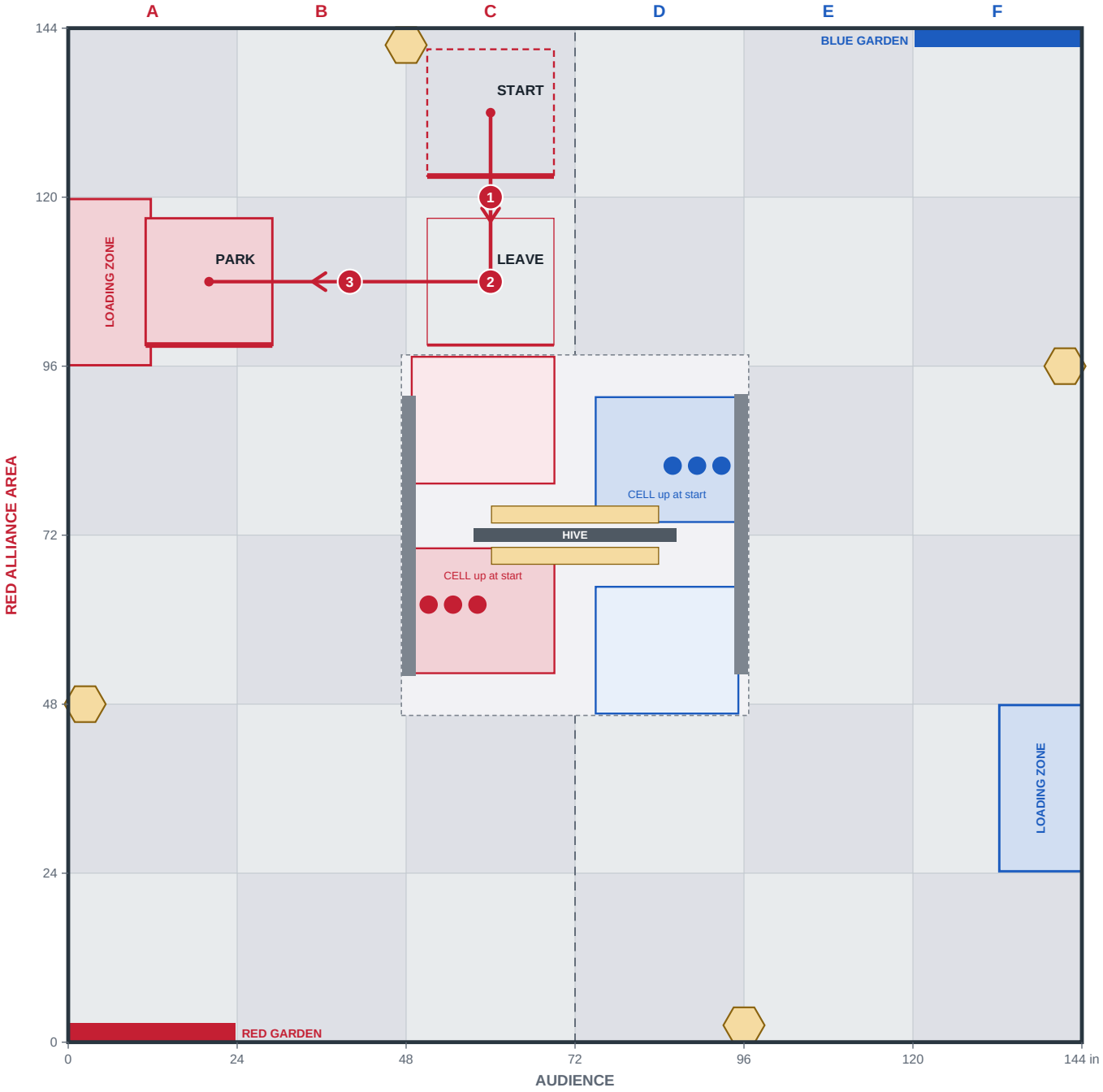
Robot: 18 x 18 in. Time per path: accelerate at 37 in/s² to 50 in/s, decelerate, then 0.25 s to settle. Action and waitUntil times aren't known (shown as ?).

Points: LEAVE 3 if the robot ends clear of every wall; PARK 5 if it ends partly in its own LOADING ZONE; HIVE tips aren't estimated until routes have launch actions.

Routes come from TeamCode/src/main/assets/autos. A copy pushed to the Control Hub (/sdcard/FIRST/autos) overrides it on the robot but isn't drawn here.

Red: Leave + Park

leave_and_park.json · route “Leave + Park” · LeaveAndParkAuto.Red · then TeleOp “Red: TeleOp” · group comp



ESTIMATED POINTS

8 points

LEAVE +3
ends 11 in from the nearest wall

PARK +5
ends partly in the red LOADING ZONE

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.7 s of 30 s

CHECKS

- START is 3.0 in from the wall; the robot must touch it (G304).

ROUTINE

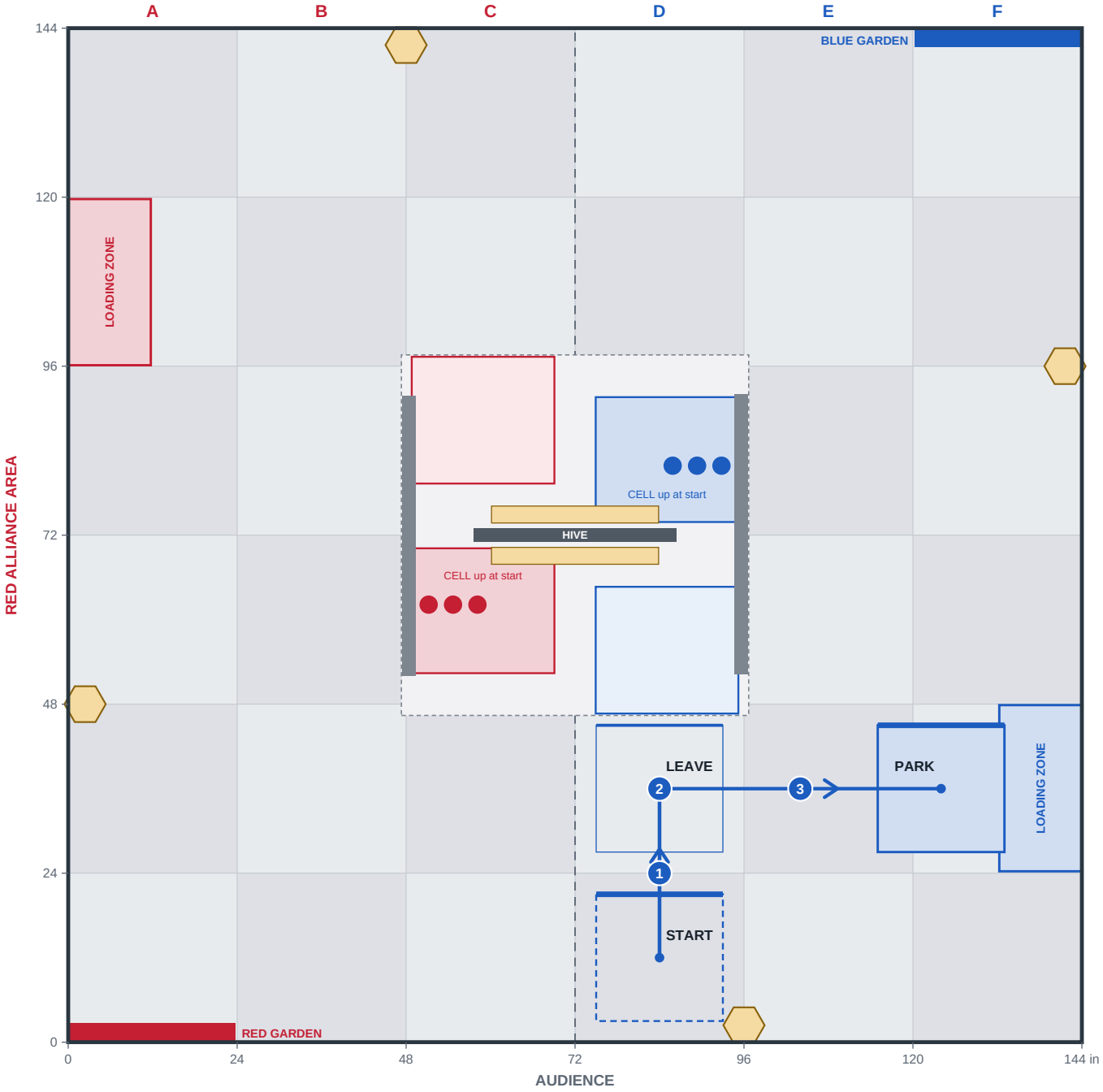
1	leave wall: line START -> LEAVE, 24 in	~1.9 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, RED FRAME (x, y, heading)

START	60.0, 132.0, 270°
LEAVE	60.0, 108.0, 270°
PARK	20.0, 108.0, 270°

Blue: Leave + Park

leave_and_park.json · route “Leave + Park” · LeaveAndParkAuto.Blue · then TeleOp “Blue: TeleOp” · group comp



ESTIMATED POINTS

8 points

LEAVE +3
ends 11 in from the nearest wall

PARK +5
ends partly in the blue LOADING ZONE

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.7 s of 30 s

CHECKS

- START is 3.0 in from the wall; the robot must touch it (G304).

ROUTINE

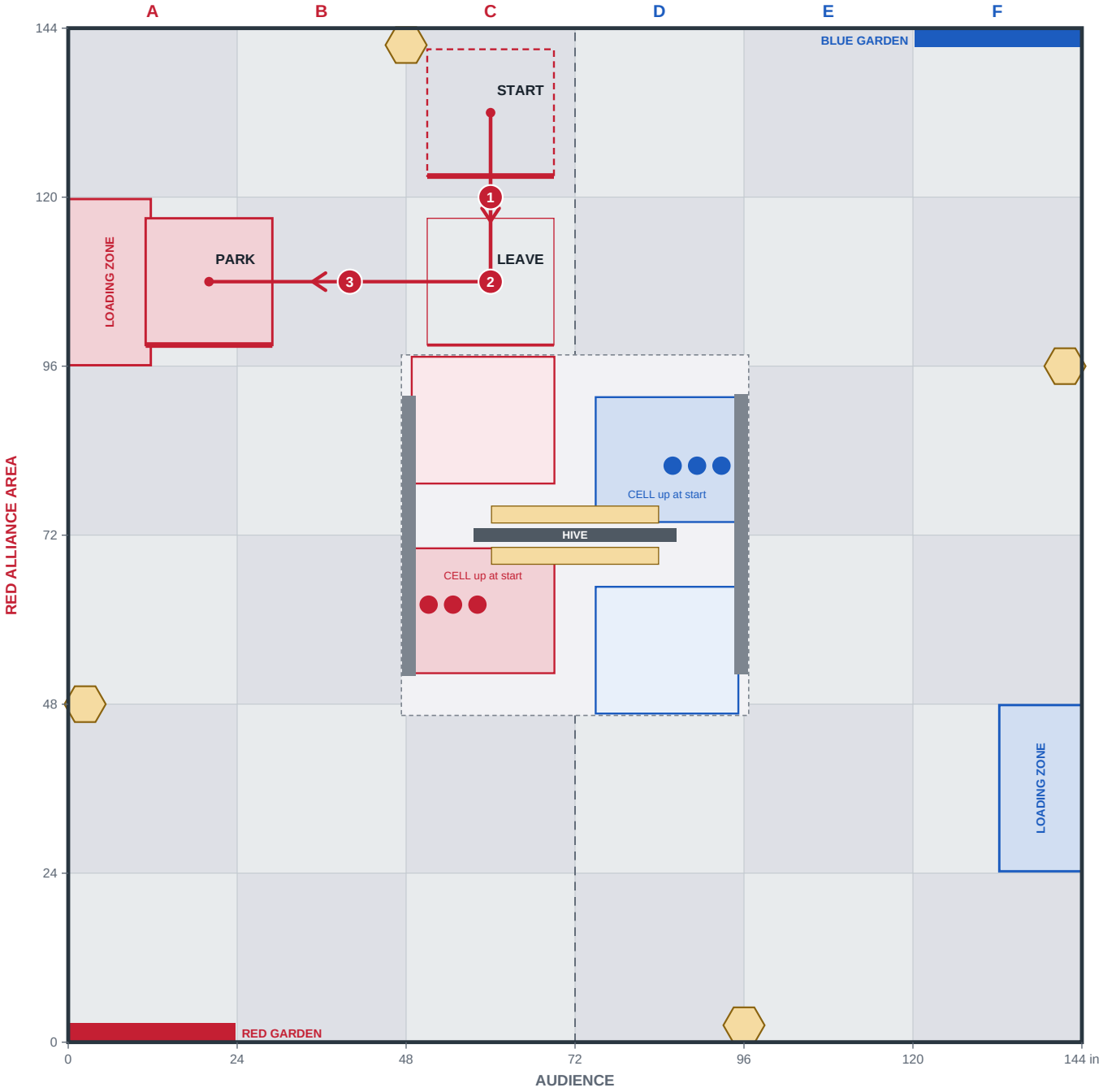
1	leave wall: line START -> LEAVE, 24 in	~1.9 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, BLUE FRAME (x, y, heading)

START	84.0, 12.0, 90°
LEAVE	84.0, 36.0, 90°
PARK	124.0, 36.0, 90°

Red: My Route

my_route.json · route “Test Route” · MyRouteAuto.Red · then TeleOp “Red: TeleOp” · group comp



ESTIMATED POINTS

8 points

LEAVE +3
ends 11 in from the nearest wall

PARK +5
ends partly in the red LOADING ZONE

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.7 s of 30 s

CHECKS

- START is 3.0 in from the wall; the robot must touch it (G304).

ROUTINE

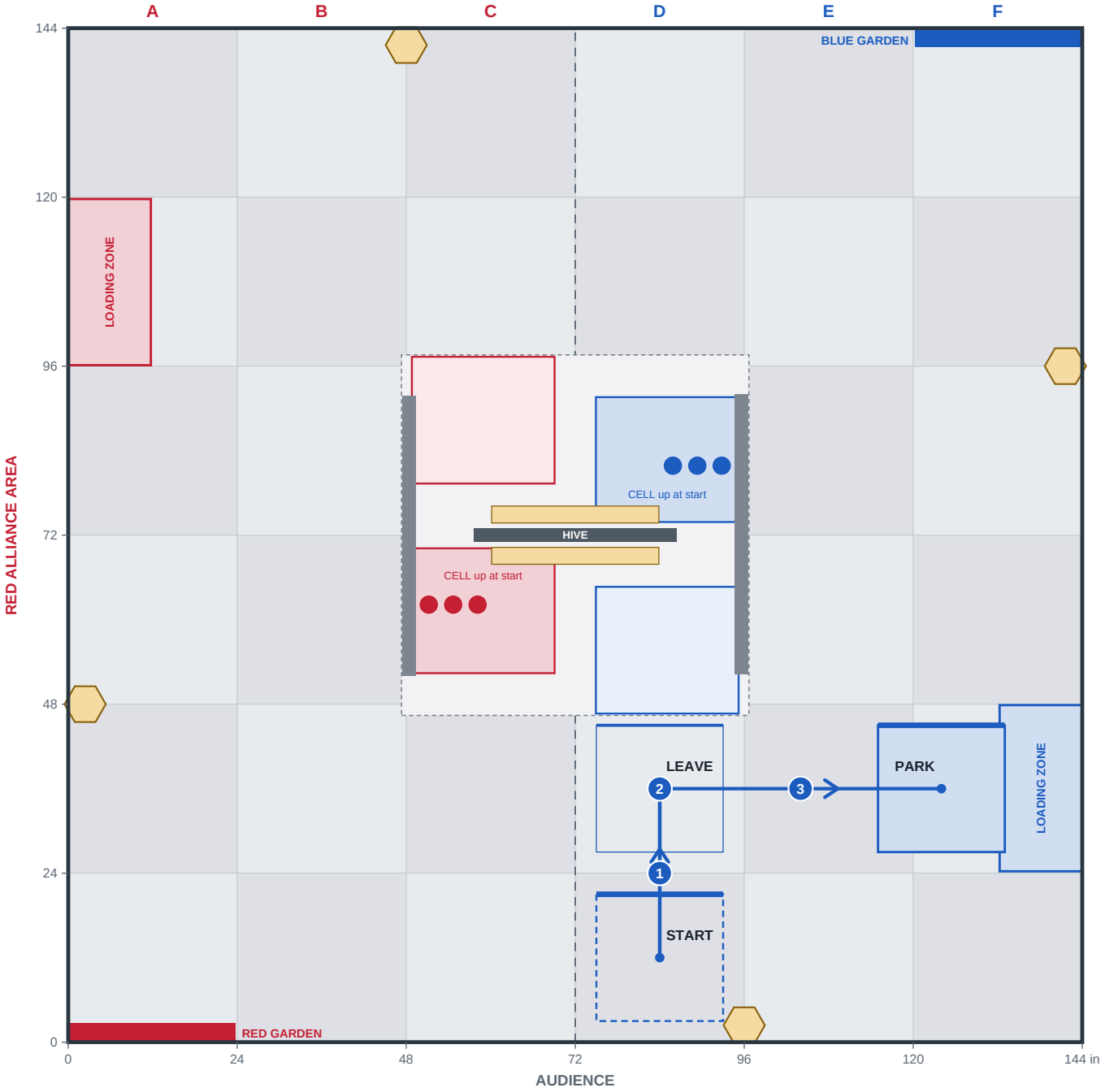
1	leave wall: line START -> LEAVE, 24 in	~1.9 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, RED FRAME (x, y, heading)

START	60.0, 132.0, 270°
LEAVE	60.0, 108.0, 270°
PARK	20.0, 108.0, 270°

Blue: My Route

my_route.json · route “Test Route” · MyRouteAuto.Blue · then TeleOp “Blue: TeleOp” · group comp



ESTIMATED POINTS

8 points

LEAVE +3
ends 11 in from the nearest wall

PARK +5
ends partly in the blue LOADING ZONE

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.7 s of 30 s

CHECKS

- START is 3.0 in from the wall; the robot must touch it (G304).

ROUTINE

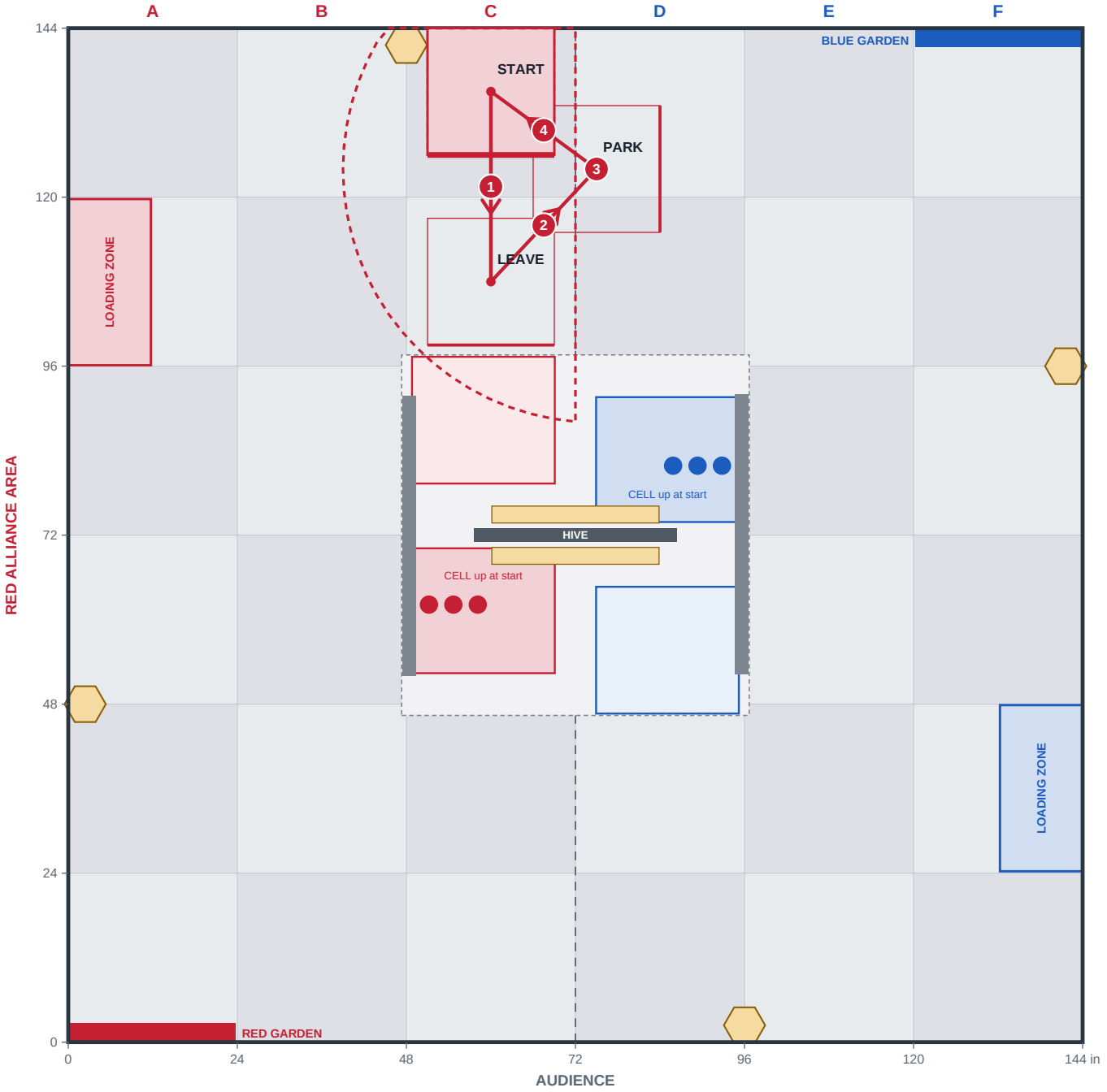
1	leave wall: line START -> LEAVE, 24 in	~1.9 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, BLUE FRAME (x, y, heading)

START	84.0, 12.0, 90°
LEAVE	84.0, 36.0, 90°
PARK	124.0, 36.0, 90°

Red: Vision Seek (test)

vision_seek.json · route “Vision Seek (test)” · VisionSeekAuto.Red · then TeleOp “Red: TeleOp” · group test



ESTIMATED POINTS

0 points

LEAVE 0
still touching the wall at the end

PARK 0
ends outside the red LOADING ZONE (x 0.0–11.7, y 96.1–119.7)

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

9.4 s of 30 s

CHECKS

- Step 4 follows a path from PARK, but after step 3 (a seek) the robot could be anywhere. Put { "driveTo": "PARK" } before it.
- Step 2 crosses into the other alliance's half (G402: risky in AUTO).
- Step 4 crosses into the other alliance's half (G402: risky in AUTO).

ROUTINE

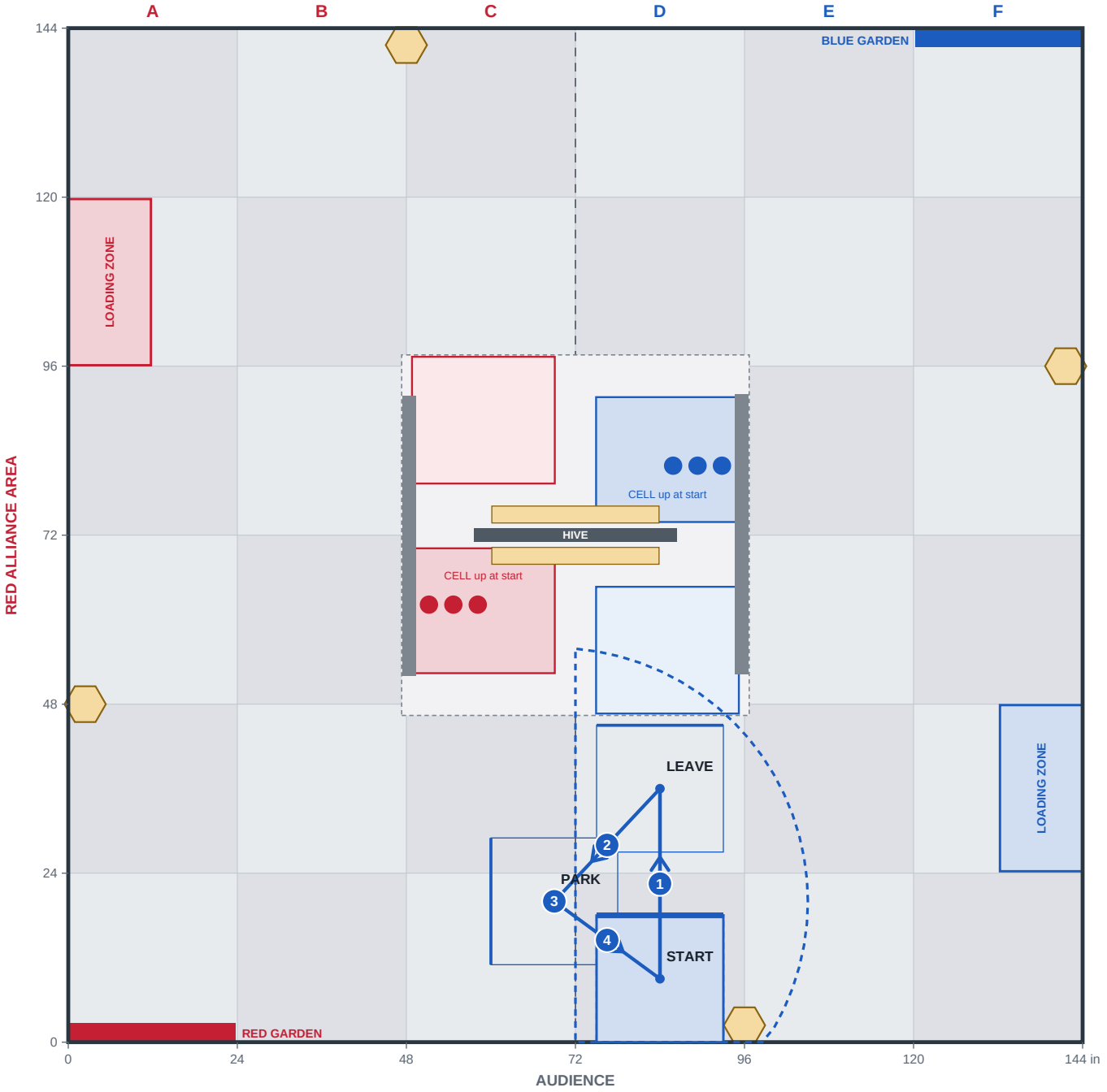
1	leave wall: line START -> LEAVE, 27 in	~2.0 s
2	line LEAVE -> PARK, 22 in	~1.8 s
3	to a POLLEN: seek vision.ball (colors pollen)	<= 4.0 s
4	line PARK -> START, 19 in	~1.7 s

WAYPOINTS, RED FRAME (x, y, heading)

START	60.0, 135.0, 270°
LEAVE	60.0, 108.0, 270°
PARK	75.0, 124.0, 0°

Blue: Vision Seek (test)

vision_seek.json · route “Vision Seek (test)” · VisionSeekAuto.Blue · then TeleOp “Blue: TeleOp” · group test



ESTIMATED POINTS

0 points

LEAVE 0
still touching the wall at the end

PARK 0
ends outside the blue LOADING ZONE (x 132.3–144.0, y 24.3–47.9)

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

9.4 s of 30 s

CHECKS

- Step 4 follows a path from PARK, but after step 3 (a seek) the robot could be anywhere. Put { "driveTo": "PARK" } before it.
- Step 2 crosses into the other alliance's half (G402: risky in AUTO).
- Step 4 crosses into the other alliance's half (G402: risky in AUTO).

ROUTINE

1	leave wall: line START -> LEAVE, 27 in	~2.0 s
2	line LEAVE -> PARK, 22 in	~1.8 s
3	to a POLLEN: seek vision.ball (colors pollen)	<= 4.0 s
4	line PARK -> START, 19 in	~1.7 s

WAYPOINTS, BLUE FRAME (x, y, heading)

START	84.0, 9.0, 90°
LEAVE	84.0, 36.0, 90°
PARK	69.0, 20.0, 180°